

5(a)	<u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s)	M1
	around a loop/around a closed circuit	A1
5(b)(i)	1. $1/R = 1/R_1 + 1/R_2$ $1/R = 1/90 + 1/18$	C1
	$R = 15\ \Omega$	A1
	2. $I = V/R$	C1
	$I = 4.8/15$ or $I = 4.8/90 + 4.8/18$ $I = 0.32\text{ A}$	A1
5(b)(ii)	$E = V + Ir$ or $E = I(R + r)$	C1
	$5.6 = 4.8 + 0.32\ r$ so $r = 2.5\ (\Omega)$ or $5.6 = 0.32 \times (15 + r)$ so $r = 2.5\ (\Omega)$	A1
5(b)(iii)	$P = EI$ or $P = VI$ or $P = I^2R$ or $P = V^2/R$	C1
	ratio = $(0.32^2 \times 2.5) / (5.6 \times 0.32)$ or $0.256 / 1.792$	C1
	= 0.14	A1

Question	Answer	Marks
5(c)	$7.2 - 5.6 - 2.5I - 3.5I = 0$	C1
	$I = 0.27 \text{ A}$	A1

Question	Answer	Marks
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